Towards a Private 5G Testbed for Manufacturing R&D

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INTRODUCTION

Private 5G Networks provide a high-speed, low-latency solution to the nearly 1.9 petabytes of data generated by manufacturing annually [1, 2]. Software Defined Radios (SDRs) offer capable security solutions and enhanced customizability, allowing flexible network configurations without hardware changes [3]. This work focuses on the analysis of a 5G Standalone (SA) Networks for future deployment within a manufacturing context with contributions in the form of:

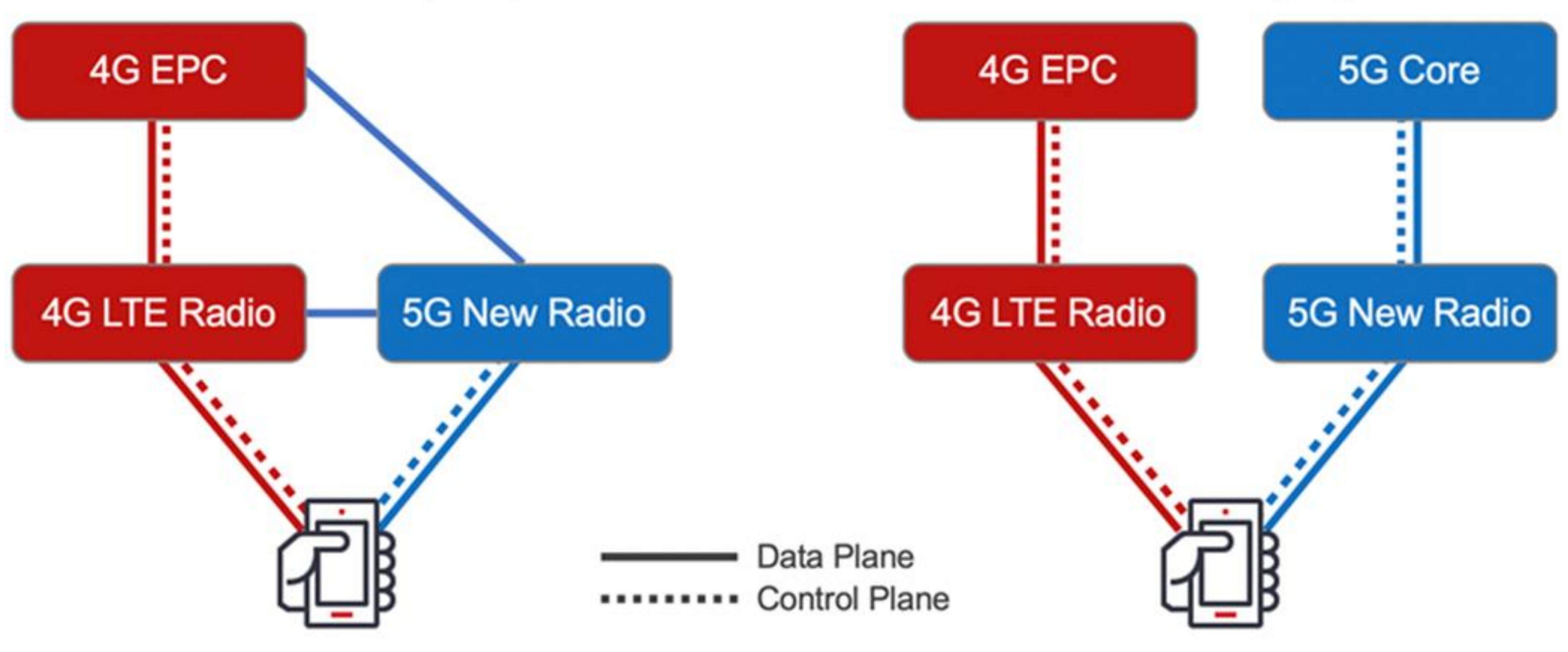
- Six configurations of OpenAirInterface [4]
- Signal strength analysis of the 5G Network
- Speed testing of the underlying 5G Network with a WiFi internet uplink
- 5G spectrum analysis under different configurations

TESTBED

CN + gNB	UE	Spectrum A nalyzer
• Dell + USRP B210 • HP + USRP B210	• USRP B210 • Raspberry Pi + SIM8202-M2	• USRP B210 • BladeRF

EXPERIMENTAL METHODOLOGY

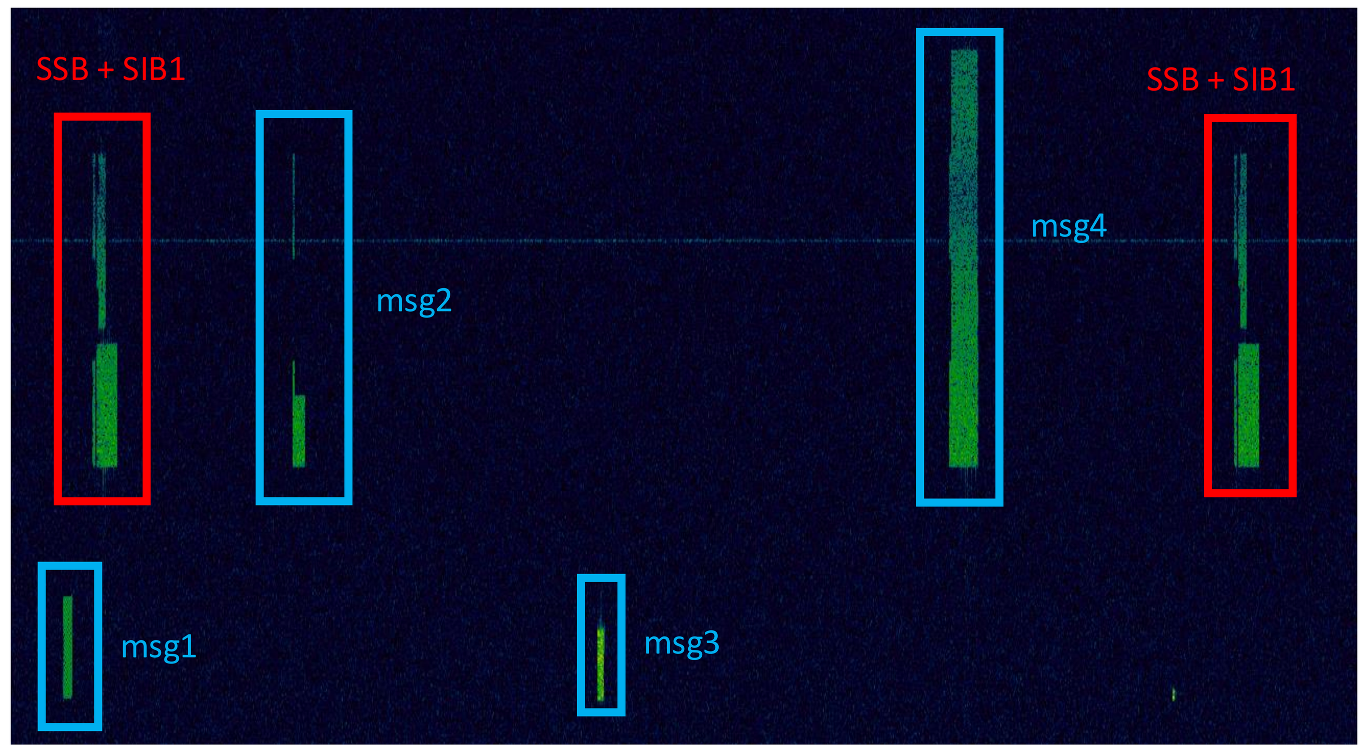
The Dell PC + USRP B210 was setup as the Core Network (CN) + gNodeB (gNB) and the Raspberry Pi was used as the User Equipment (UE) in each experiment. The spectrum analysis was performed using the signals that were captured using the USRP. The average Reference Signal Received Power (RSRP) was measured from the gNB to create the heat map. Fast.com and Speedtest.net were used to measure the speed of the WiFi on the Dell alone and on the Raspberry Pi when connected to the Dell over 5G.



5G NSA vs. 5G SA [6]

RESULTS

5G Spectrum Analysis



Annotated figure of the 5G Random Access Channel (RACH) Procedure

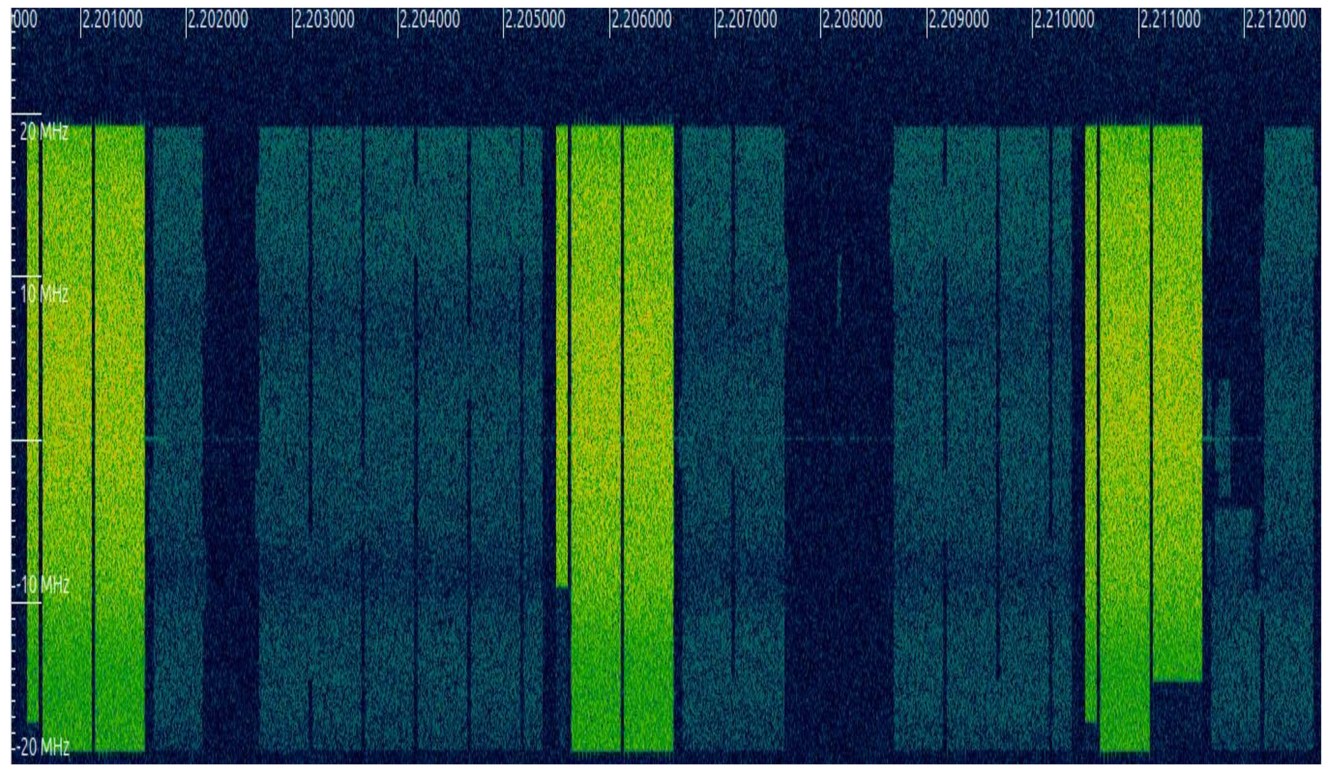
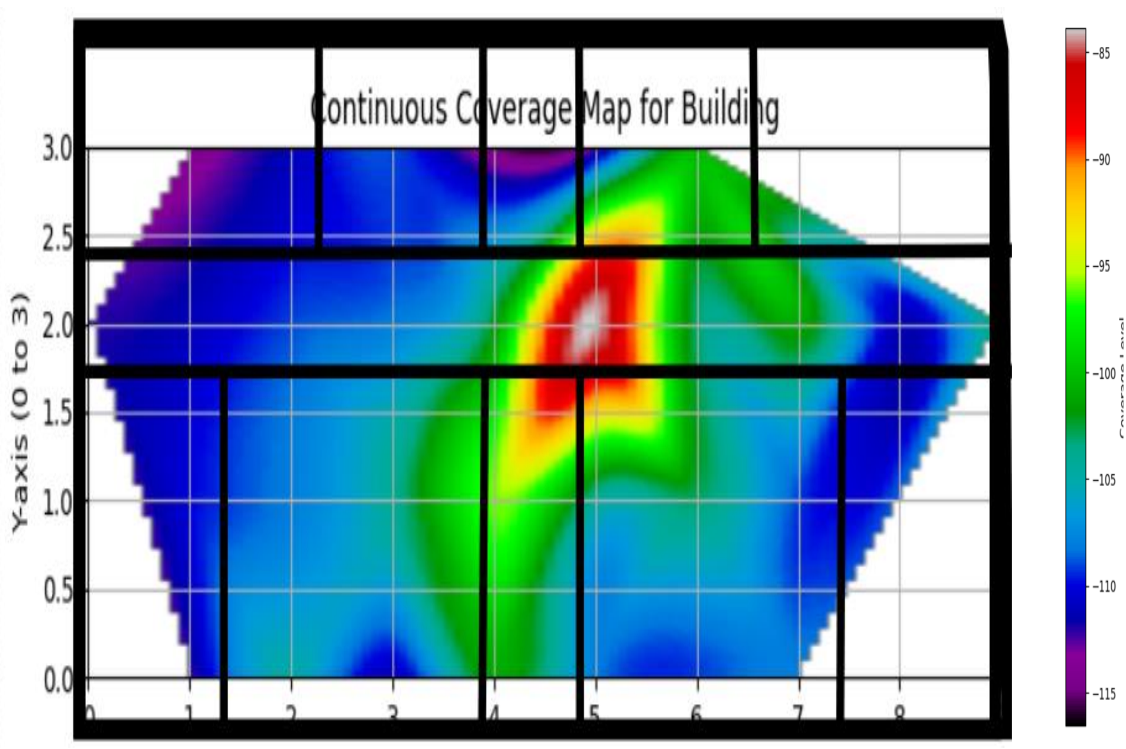


Figure showing high-throughput data transfer over 5G

Signal Strength Mapping



Throughput Analysis

Metrics	Speedtest. net (WiFi)	Fast.com (WiFi)	Speedtest.net (WiFi + 5G)	Fast.com (WiFi + 5G)
Download speed	61.33 Mbps	72.5 Mbps	27.17 Mbps	7.66 Mbps
Upload speed	78.88 Mbps	89 Mbps	13.89 Mbps	13.3 Mbps

DISCUSSION

The findings presented in this work show the capabilities of a 5G SA network for manufacturing using currently available tools. Utilizing the OAI framework, multiple performance metrics were analyzed to determine the power of 5G Networks for manufacturing solutions. The data shows promising results for the future of 5G, while highlighting the improvements that stand to be made towards implementation in manufacturing contexts.

Future work for this project entails scaling up the testbed toward an O-RAN split 7.2 implementation in addition to developing continuous integration/continuous development of the testbed as a whole and integration into manufacturing-specific applications.

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